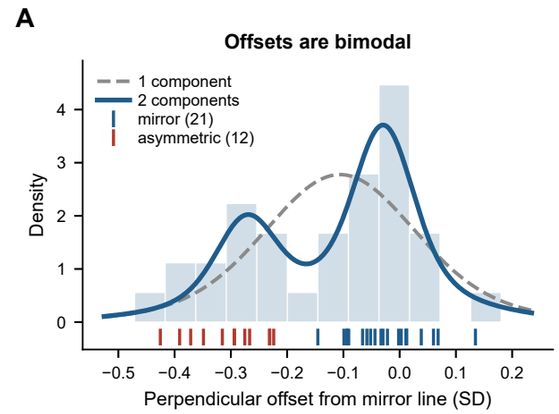
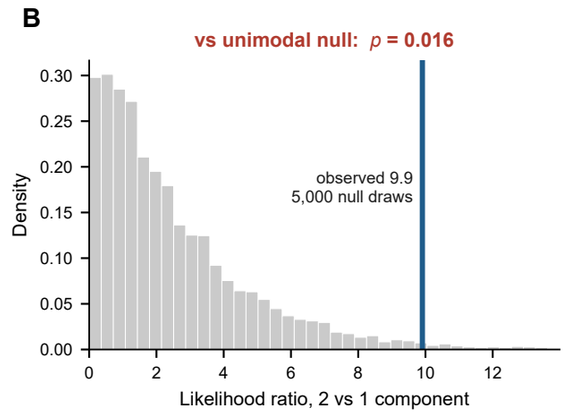
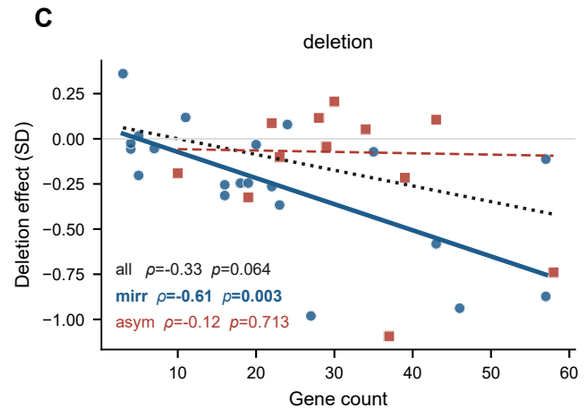
Asymmetry is a rare single hit on an oligogenic mirror background

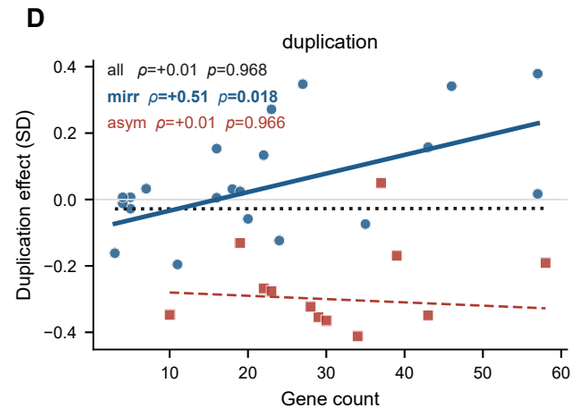
1. The perpendicular offset from the mirror line is bimodal



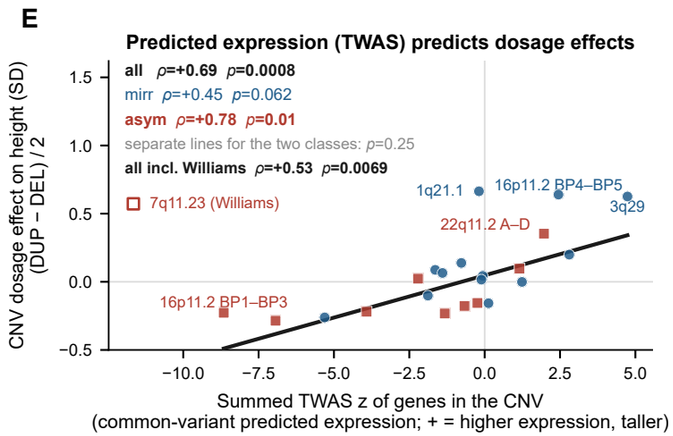


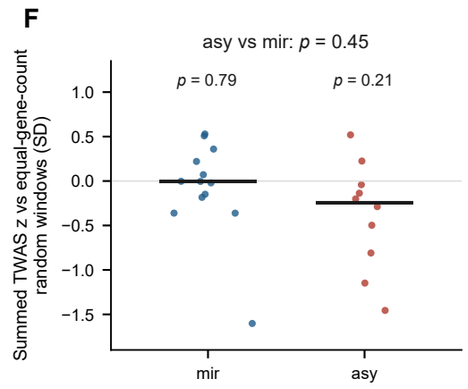
2. Effect size scales with gene count only after stratifying by class



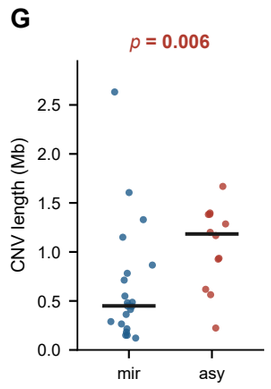


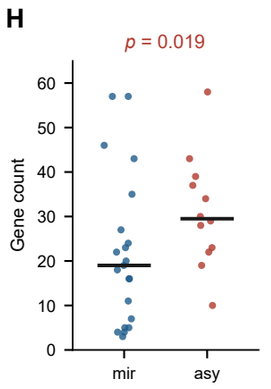
3. Common-variant gene effects predict CNV dosage effects

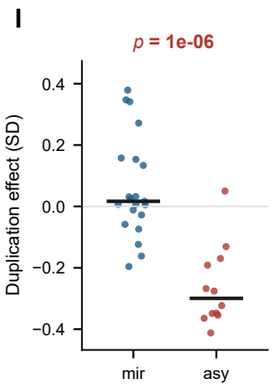


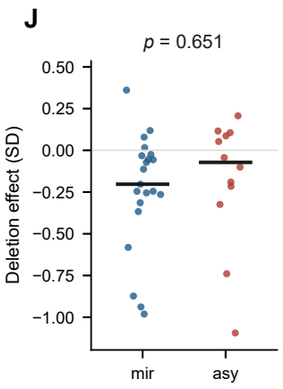


4. Mirror and asymmetric CNVs differ by size and DUP effect, not DEL

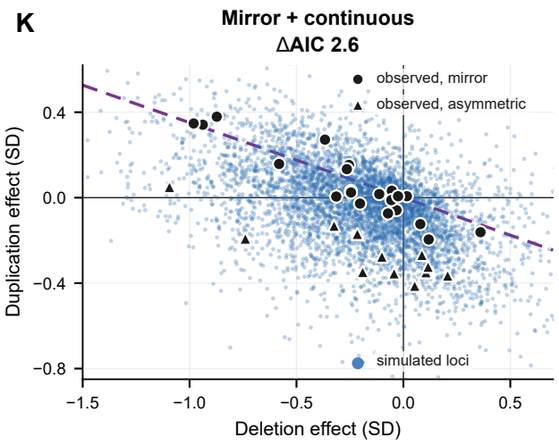


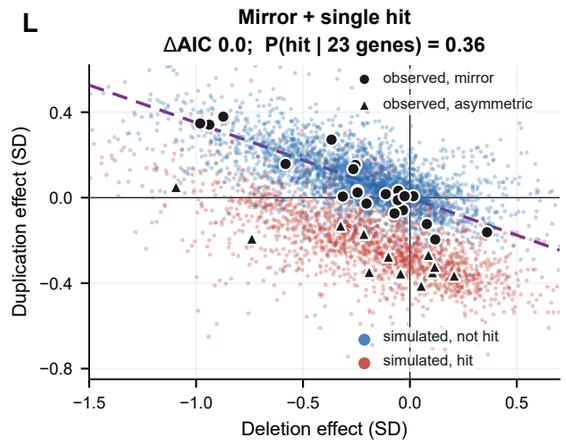




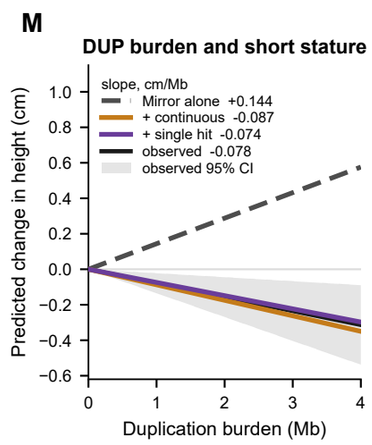


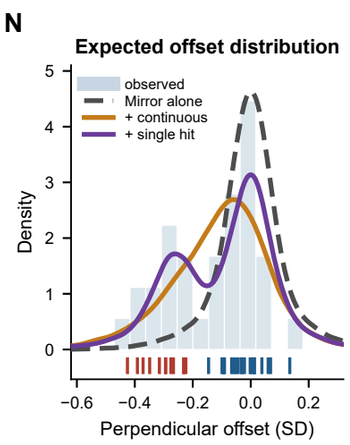
5. Two-component (mirror + asymmetric) models

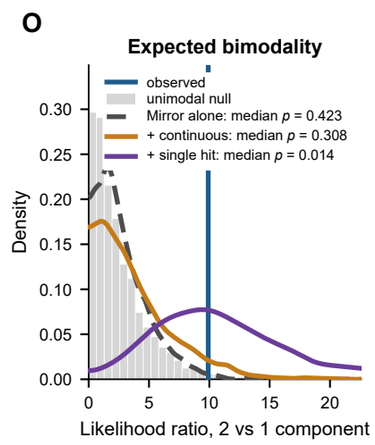




6. Where the models differ: duplication burden and bimodality







M: each model is calibrated so its deletion burden equals the published -0.41 cm/Mb (Macé et al. 2017); its duplication slope is then an out-of-sample prediction. Grey band = published 95% CI.
N, O: each model's 2,000 simulated datasets (real gene counts and SEs). N pools their offsets; O shows their likelihood ratios. Median *p* = the typical dataset's *p* against the unimodal null in B.
E: dosage effect = (DUP − DEL)/2, which cancels a shift shared by both alleles and halves one on the duplication axis; TWAS z summed over the genes in the CNV (Mancuso et al. 2017); open square = Williams–Beuren region (7q11.23), excluded from the Theil–Sen line and the correlations; *p* from circular genomic permutation. F: each CNV's summed z relative to random windows with the same number of genes; bars = medians. Non-overlapping CNVs only.